

WEB SCRAPING

Data Collection Using Python and BeautifulSoup

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OBJECTIVE

- Extract data from a public website.
- Learn web scraping using Python.
- Parse HTML content using BeautifulSoup.
- Create a structured dataset for analysis.

TOOLS AND TECHNOLOGIES

- Python
- Requests Library
- BeautifulSoup
- Pandas
- CSV File Format

METHODOLOGY

- 1. Select a website for scraping.
- 2. Send a request to the website.
- 3. Retrieve HTML content.
- 4. Parse HTML using BeautifulSoup.
- 5. Extract required information.
- 6. Store data in a CSV file.

PYTHON IMPLEMENTATION

```
import requests
from bs4 import BeautifulSoup
import pandas as pd

# Website to scrape
url = "https://books.toscrape.com/"

# Send request
response = requests.get(url)

# Parse HTML
soup = BeautifulSoup(response.text, "html.parser")

# Extract book titles
books = soup.find_all("h3")

titles = []

for book in books:
    title = book.find("a")["title"]
    titles.append(title)

# Create dataframe
df = pd.DataFrame(titles, columns=["Book Title"])

# Save dataset
df.to_csv("books_dataset.csv", index=False)

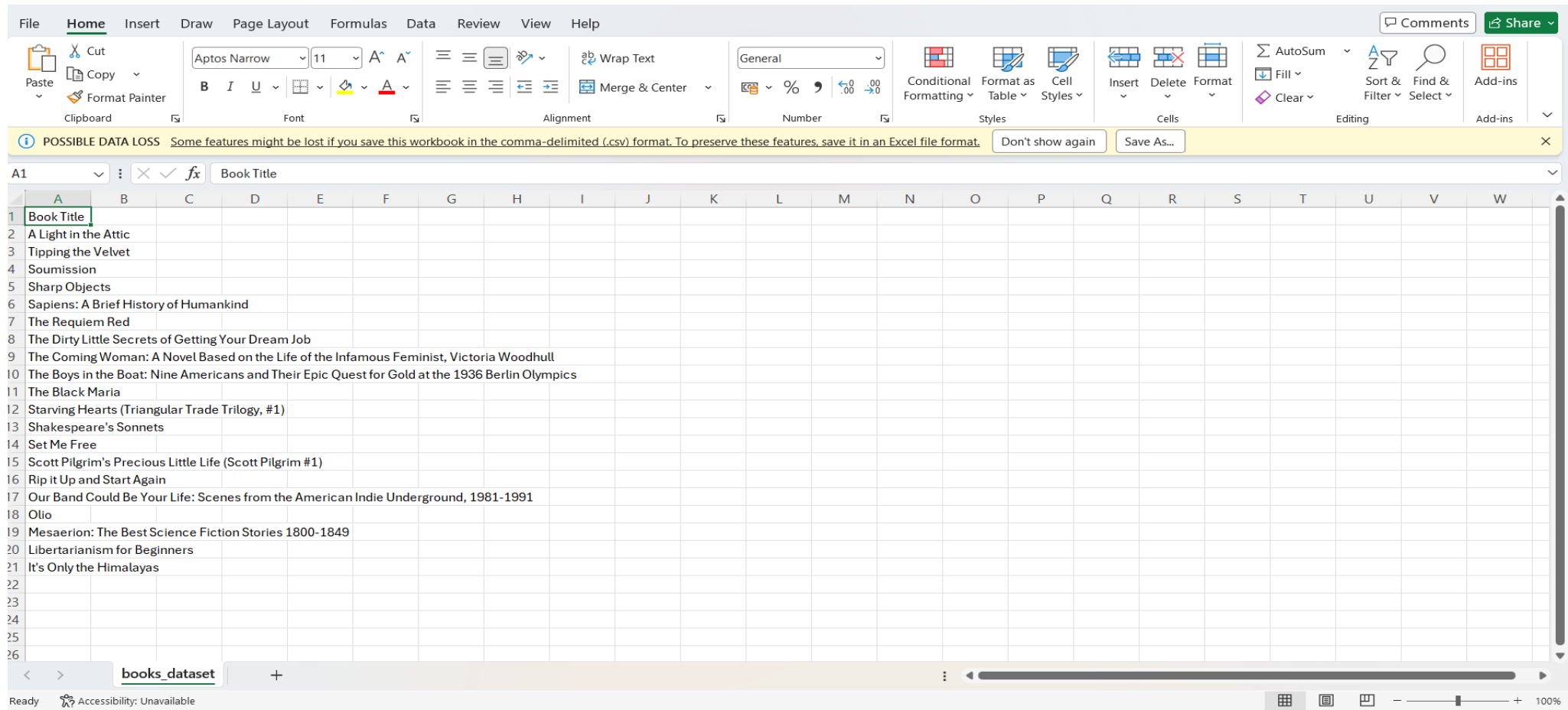
print(df.head())
print("Dataset saved successfully!")
```

PROGRAM OUTPUT

```
Python 3.14.6 (tags/v3.14.6:c63aec6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit
(AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/Dhivya/Documents/task1_webscraping.py =====
                Book Title
0                A Light in the Attic
1                Tipping the Velvet
2                Soumission
3                Sharp Objects
4  Sapiens: A Brief History of Humankind
Dataset saved successfully!
```

DATASET CREATED

Output File: books_dataset.csv



RESULTS

- Successfully extracted book titles.
- Created a structured dataset.
- Automated data collection process.
- Prepared data for future analysis.

CONCLUSION

- Learned web scraping fundamentals.
- Understood HTML parsing techniques.
- Used Python libraries for automation.
- Generated a reusable dataset.

THANKYOU